

E9 205 Machine Learning for Signal Processing

Mixture Gaussian Modeling and EM Algorithm

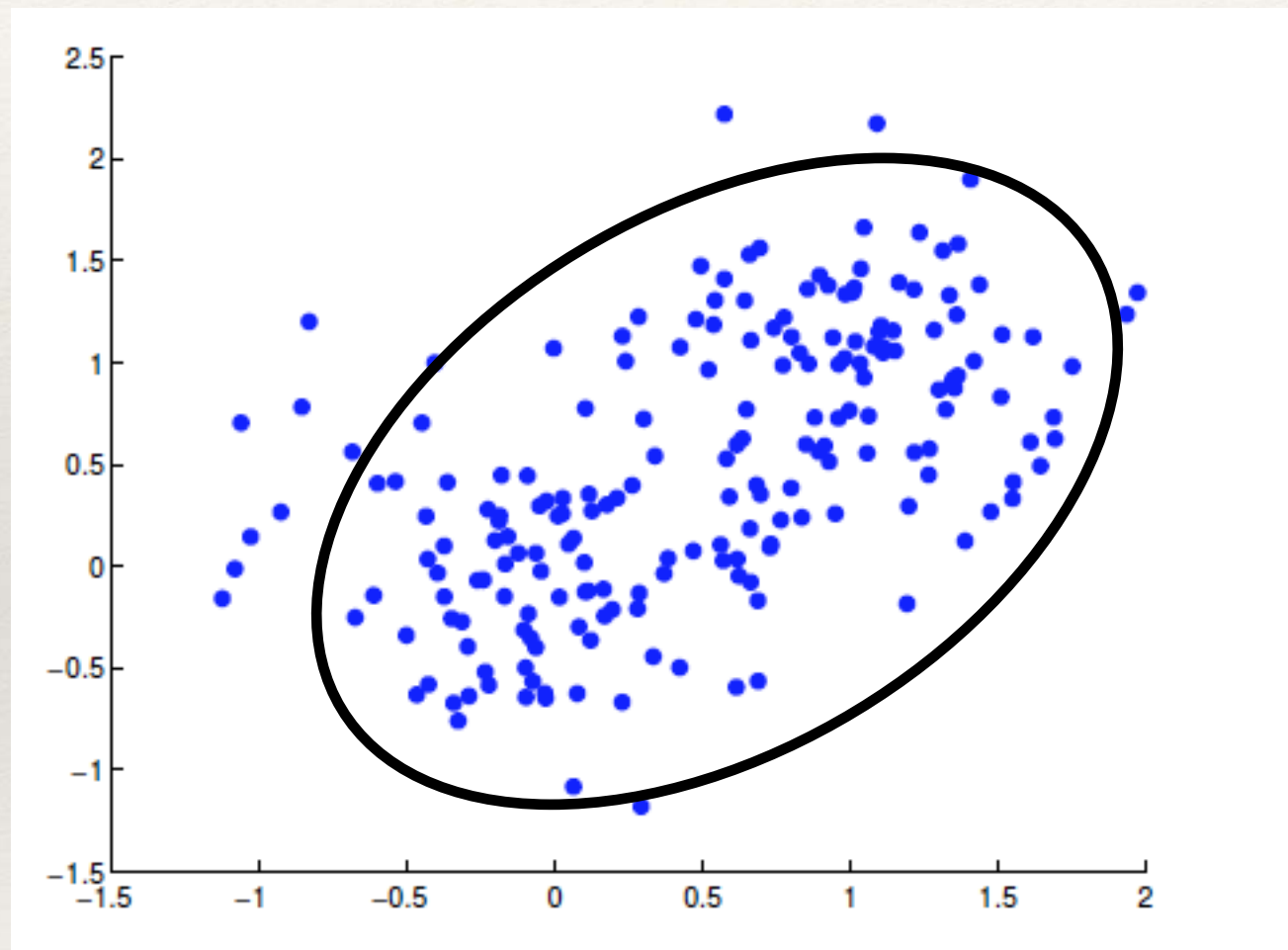
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Gaussian Distribution Summary

- ❖ The Gaussian model - parametric distributions
- ❖ Simple and useful properties.
- ❖ Can model unimodal (single peak distributions)
- ❖ MLE gives intuitive results
- ❖ Issues with Gaussian model
 - ❖ Multi-modal data
 - ❖ Not useful for complex data distributions
- ❖ Need for mixture models

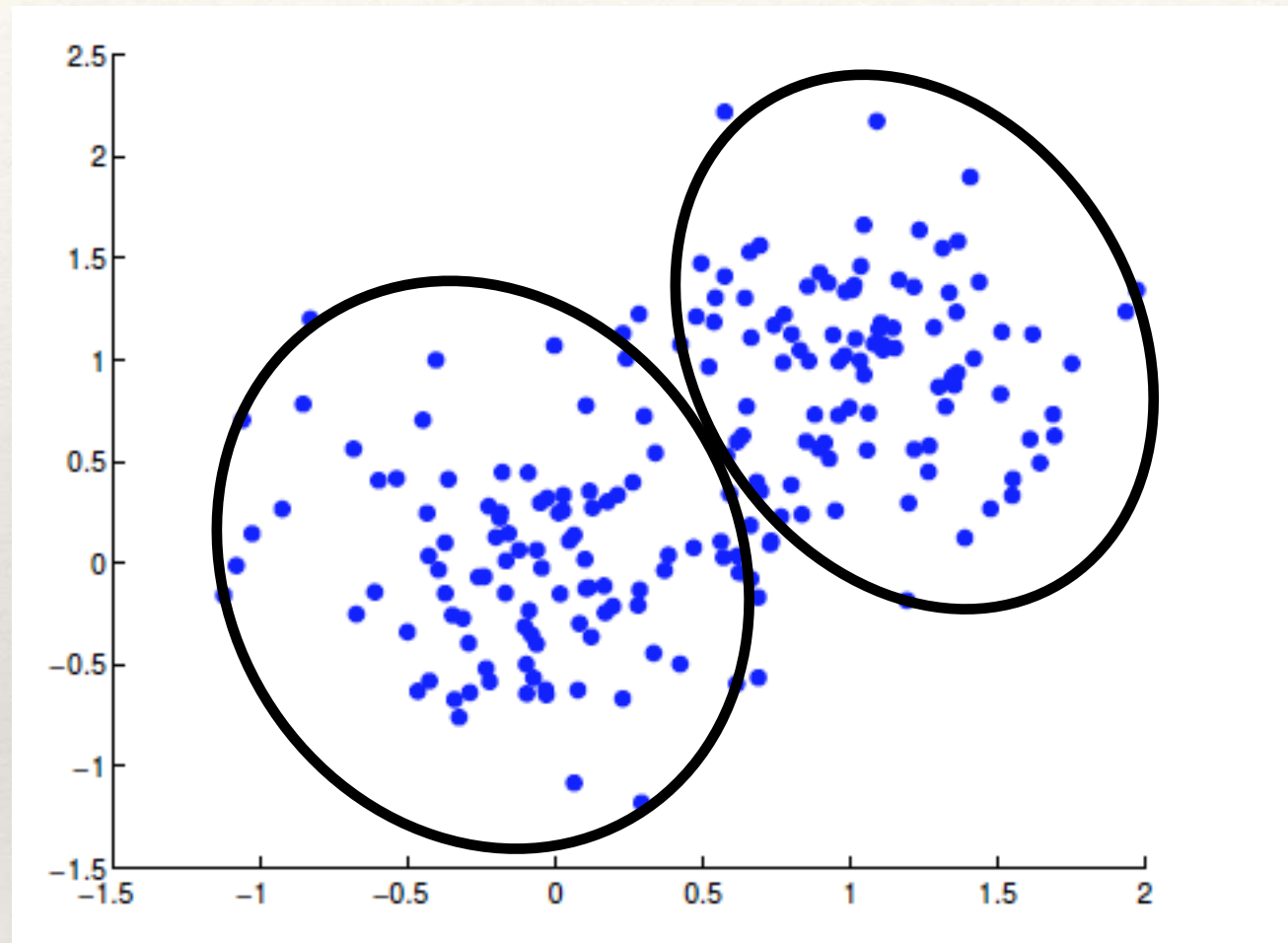
Gaussian Distribution

Often the data lies in clusters (2-D example)



Fitting a single Gaussian model may be **too broad**.

Gaussian Distribution

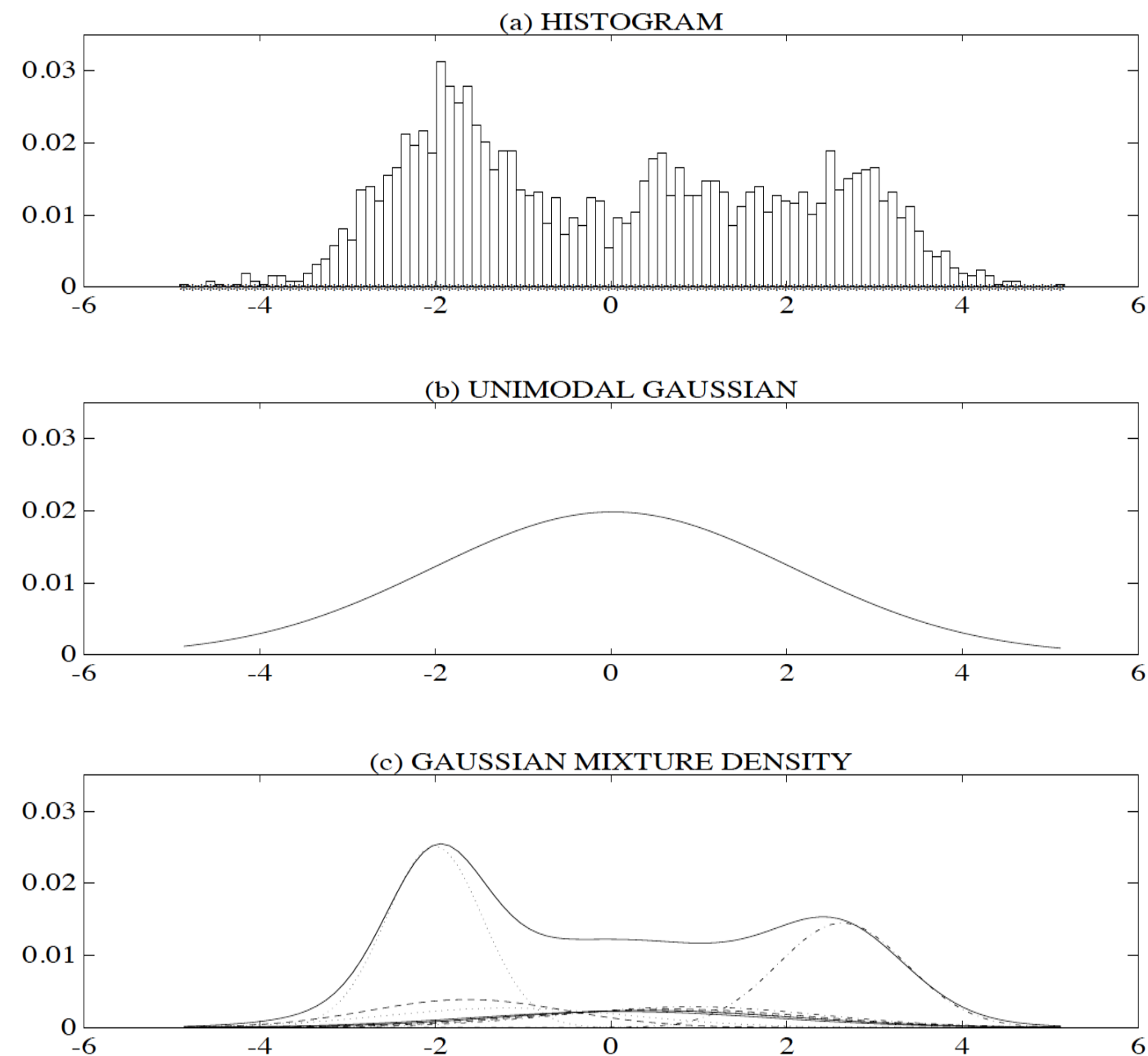


Need mixture models

Can fit any arbitrary distribution.

Gaussian Distribution

1-D example



Often the data lies in clusters

Gaussian Mixture Models

A Gaussian Mixture Model (GMM) is defined as

$$p(\mathbf{x}|\Theta) = \sum_{k=1}^K \alpha_k p(\mathbf{x}|\theta_k)$$

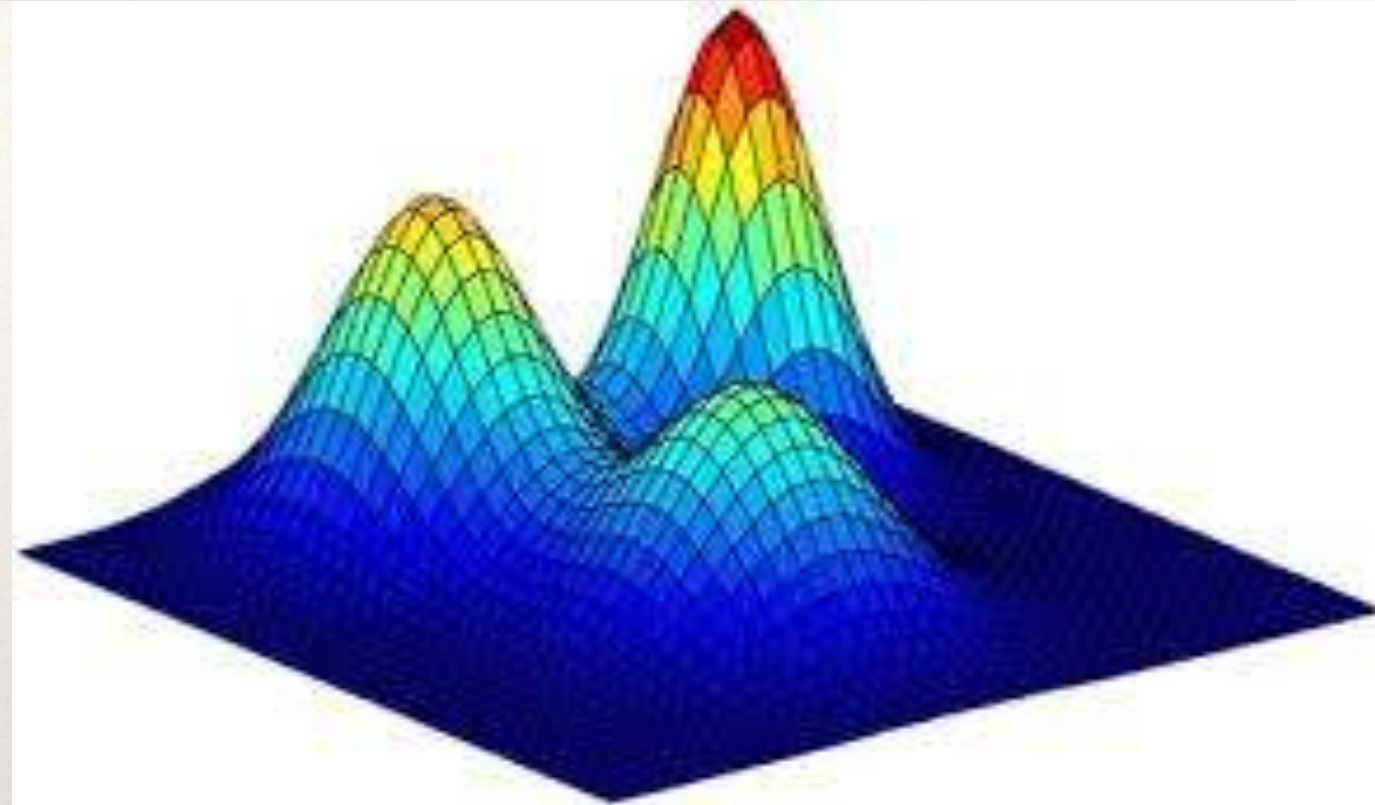
$$p(\mathbf{x}|\theta_k) = \frac{1}{\sqrt{(2\pi)^D |\Sigma_k|}} \exp\left\{ -\frac{1}{2}(\mathbf{x} - \mu_k)^* \Sigma_k^{-1} (\mathbf{x} - \mu_k) \right\}$$

The weighting coefficients have the property

$$\sum_{k=1}^K \alpha_k = 1$$

Gaussian Mixture Models

- ❖ Properties of GMM
 - ❖ Can model multi-modal data.
 - ❖ Identify data clusters.
 - ❖ Can model arbitrarily complex data distributions



The set of parameters for the model are

$$\Theta = \{\alpha_k, \theta_k\}_{k=1}^K \quad \theta_k = \{\mu_k, \Sigma_k\}$$

The number of parameters is $KD^2 + KD + K$

MLE for GMM

- ❖ The log-likelihood function over the entire data in this case will have a **logarithm of a summation**

$$\log L(\Theta) = \sum_{i=1}^N \log \left(\sum_{k=1}^K \alpha_k p(\mathbf{x}_i | \theta_k) \right)$$

- ❖ Solving for the optimal parameters using MLE for GMM is not straight forward.
- ❖ Resort to the **Expectation Maximization (EM)** algorithm

Expectation Maximization Algorithm

- ❖ Iterative procedure.
- ❖ Assume the existence of hidden variable associated with each data sample
- ❖ Let the current estimate (at iteration n) be Θ^n . Define the Q function as

$$\begin{aligned} Q(\Theta, \Theta^n) &= E_{\mathbf{z}|\mathbf{X}, \Theta^n} [\log(P(\mathbf{X}, \mathbf{z}|\Theta))] \\ &= \sum_{\mathbf{z}} \log(P(\mathbf{X}, \mathbf{z}|\Theta)) P(\mathbf{z}|\mathbf{X}, \Theta^n) \end{aligned}$$

Expectation Maximization Algorithm

- ❖ It can be proven that if we choose

$$\Theta^{n+1} = \arg \max_{\Theta} Q(\Theta, \Theta^n)$$

then $L(\Theta^{n+1}) \geq L(\Theta^n)$

- ❖ In many cases, finding the maximum for the Q function **may be easier** than likelihood function w.r.t. the parameters.
- ❖ Solution is dependent on finding **a good choice of the hidden variables** which eases the computation
- ❖ **Iteratively** improve the log-likelihood function.

EM Algorithm Summary

- ❖ Initialize with a set of model parameters ($n=1$)
- ❖ Compute the conditional expectation (E-step)

$$E_{\mathbf{z}|\mathbf{X},\Theta^n} [\log(P(\mathbf{X}, \mathbf{z}|\Theta))]$$

- ❖ Maximize the conditional expectation w.r.t. parameter. (M-step) ($n = n+1$)
- ❖ Check for convergence
- ❖ Go back to E-step if model has not converged.

EM Algorithm for GMM

- ❖ The hidden variables $\mathbf{z}_i = l$ will be the index of the mixture component which generated $\mathbf{x}_i$
- ❖ Re-estimation formulae

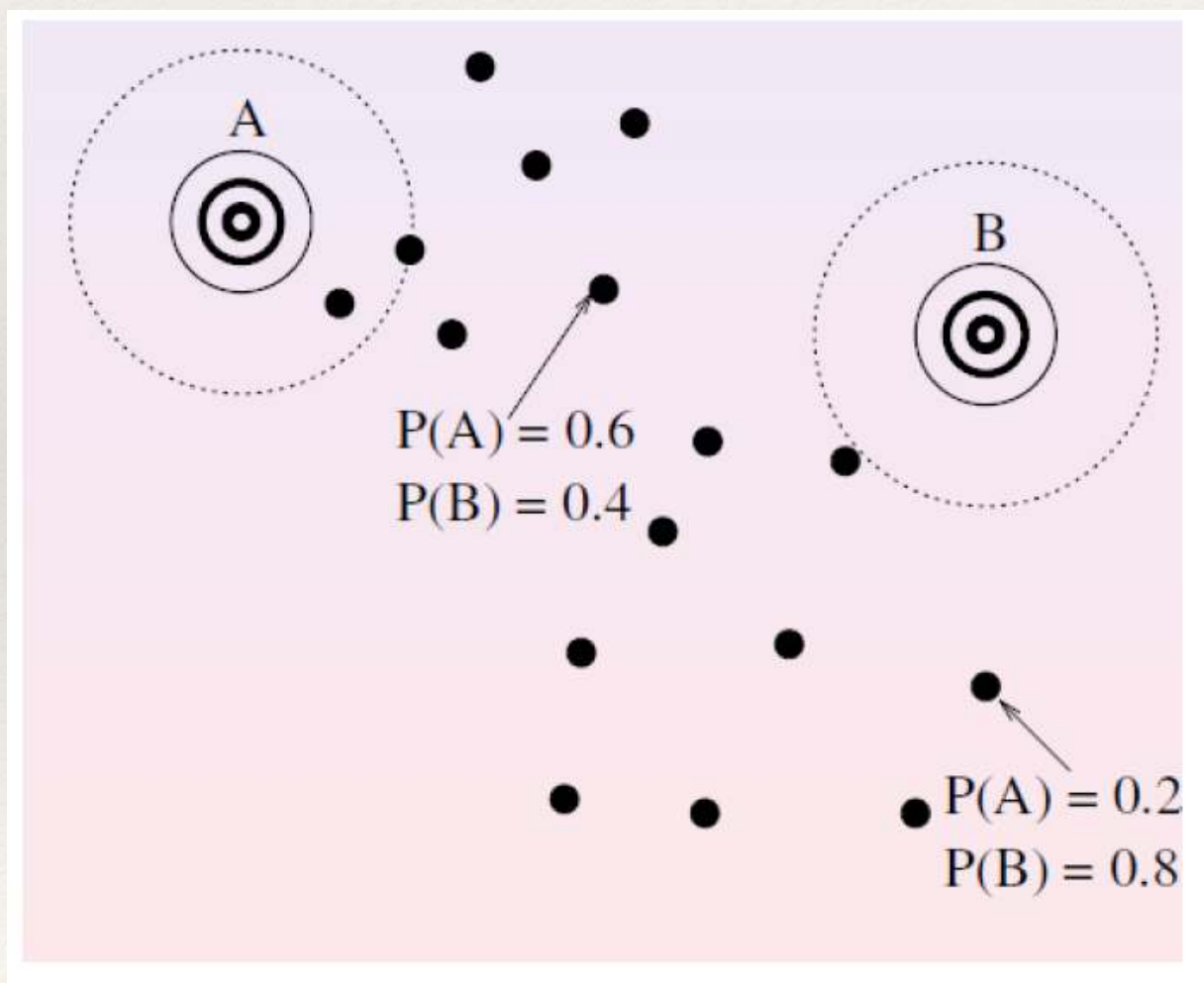
$$\alpha_{\ell}^{new} = \frac{1}{N} \sum_{i=1}^N p(\ell | x_i, \Theta^g)$$

$$\mu_{\ell}^{new} = \frac{\sum_{i=1}^N x_i p(\ell | x_i, \Theta^g)}{\sum_{i=1}^N p(\ell | x_i, \Theta^g)}$$

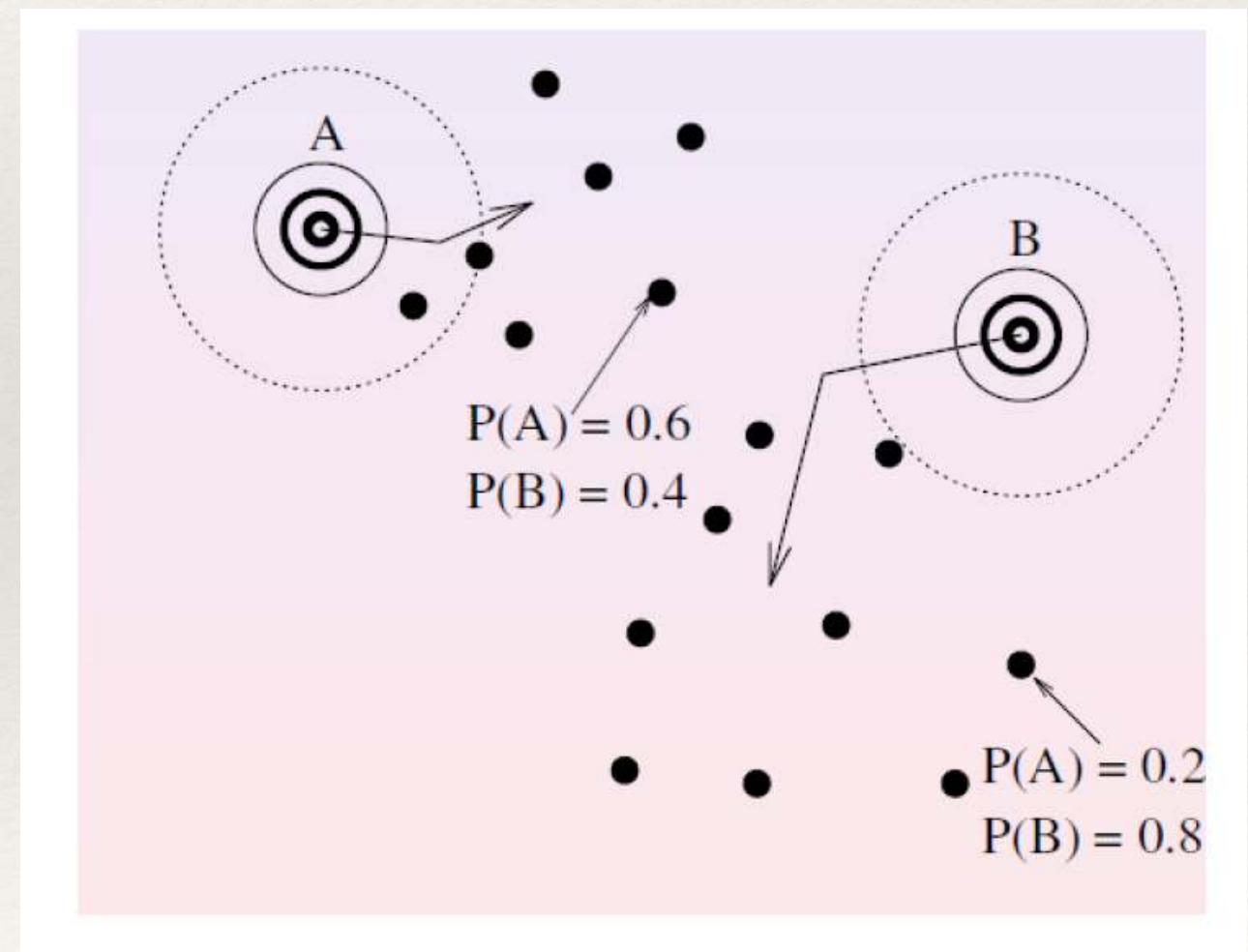
$$\Sigma_{\ell}^{new} = \frac{\sum_{i=1}^N p(\ell | x_i, \Theta^g) (x_i - \mu_{\ell}^{new})(x_i - \mu_{\ell}^{new})^T}{\sum_{i=1}^N p(\ell | x_i, \Theta^g)}$$

EM Algorithm for GMM

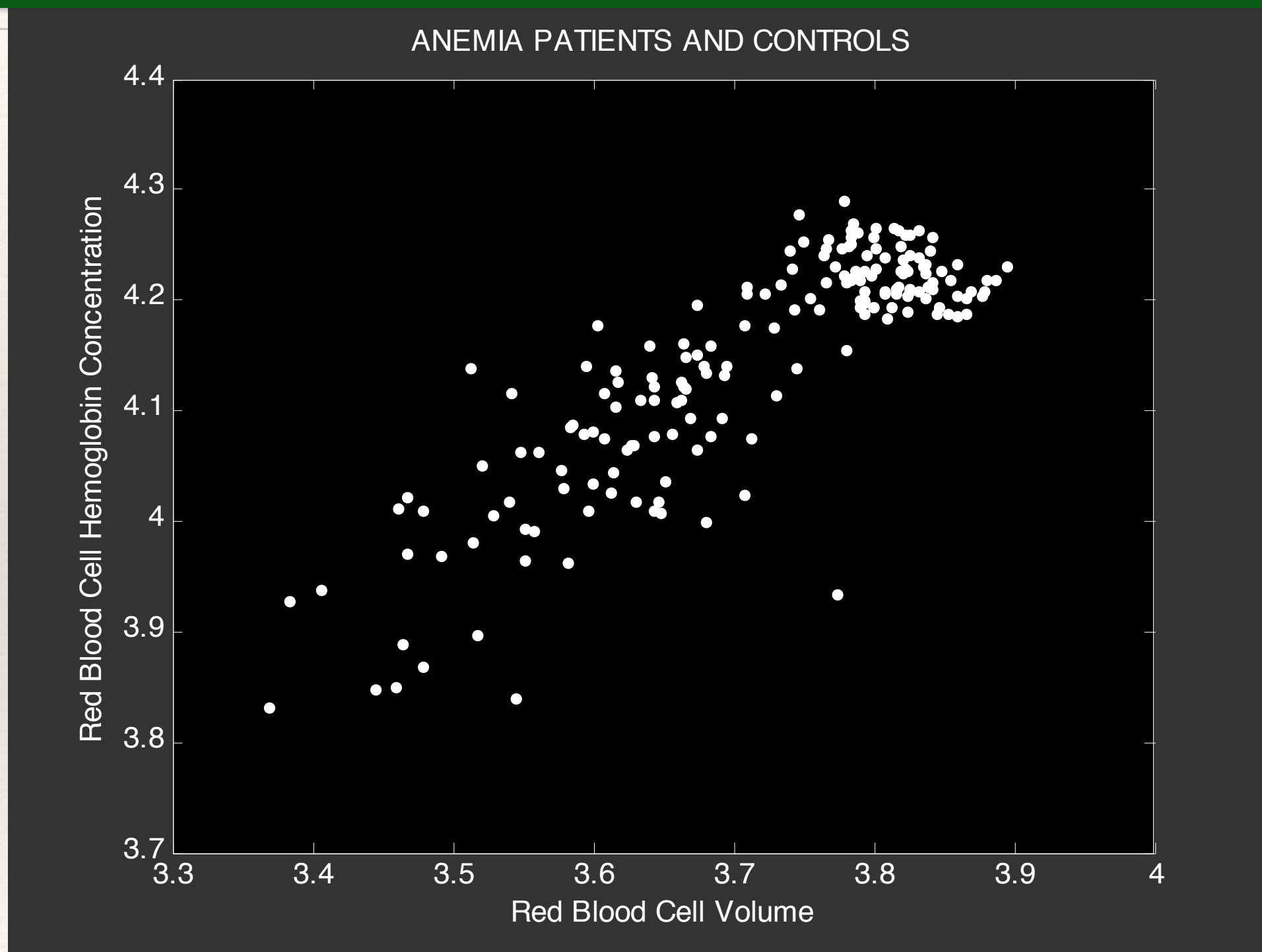
E-step



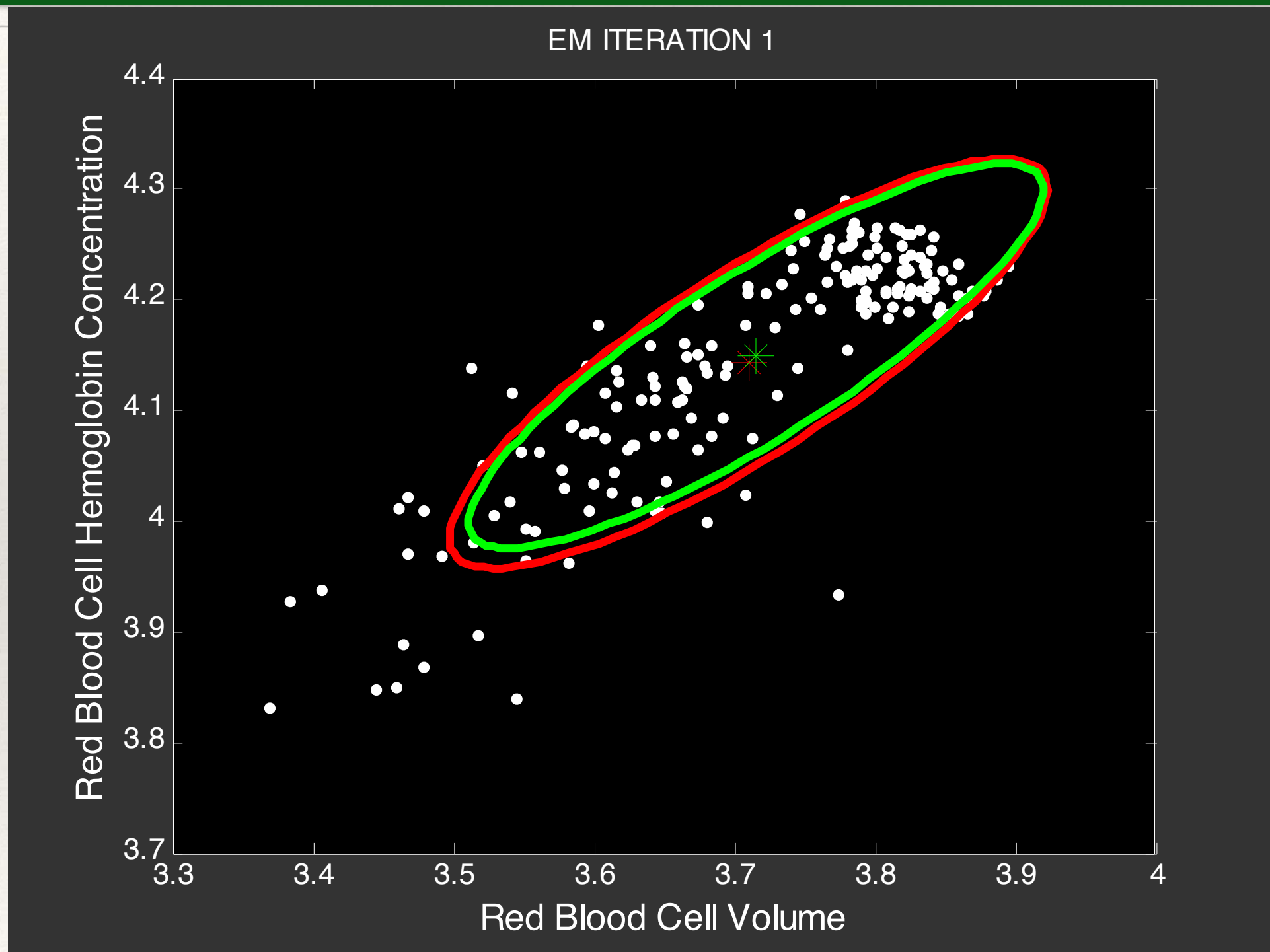
M-step



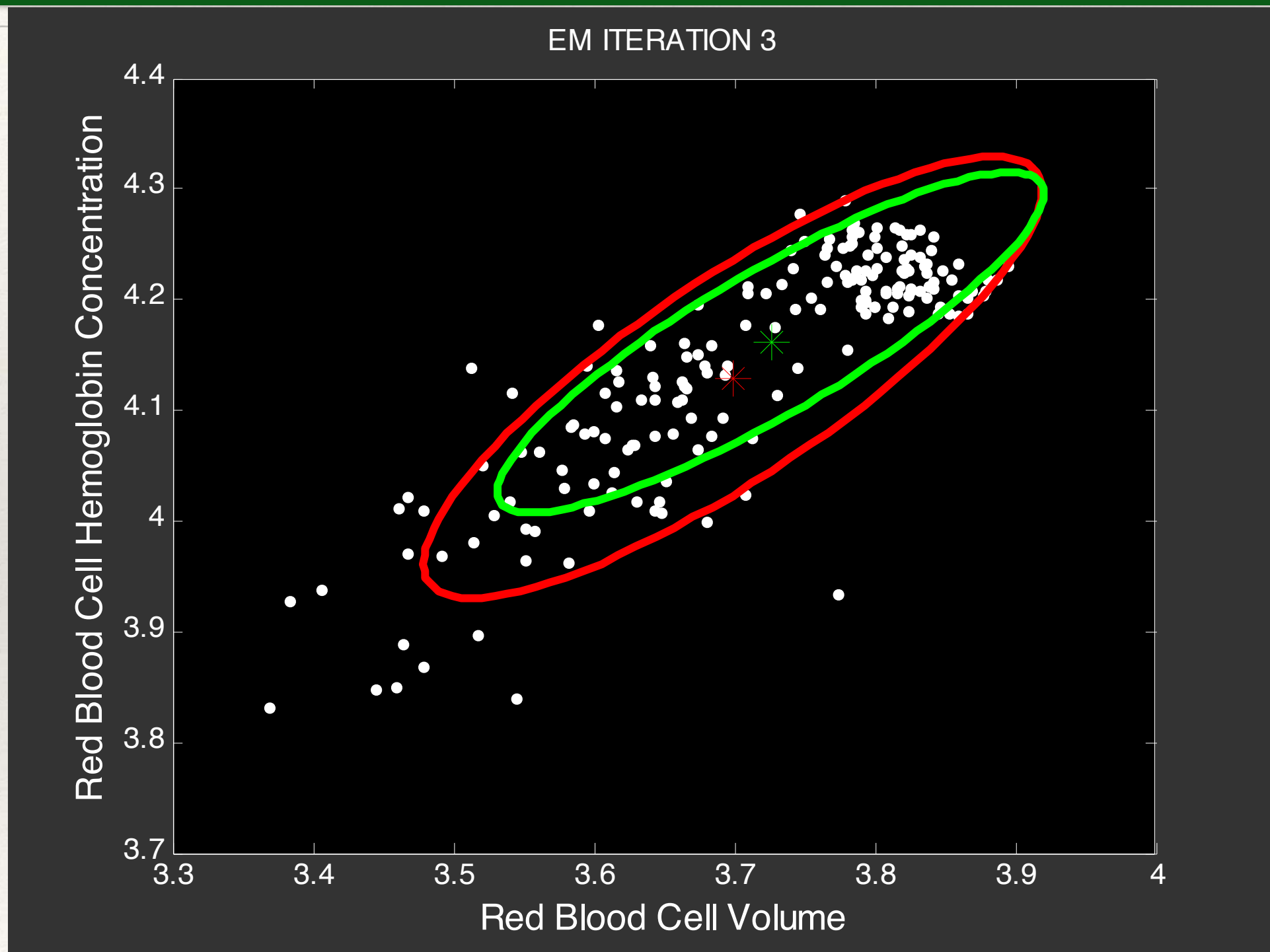
EM Algorithm for GMM



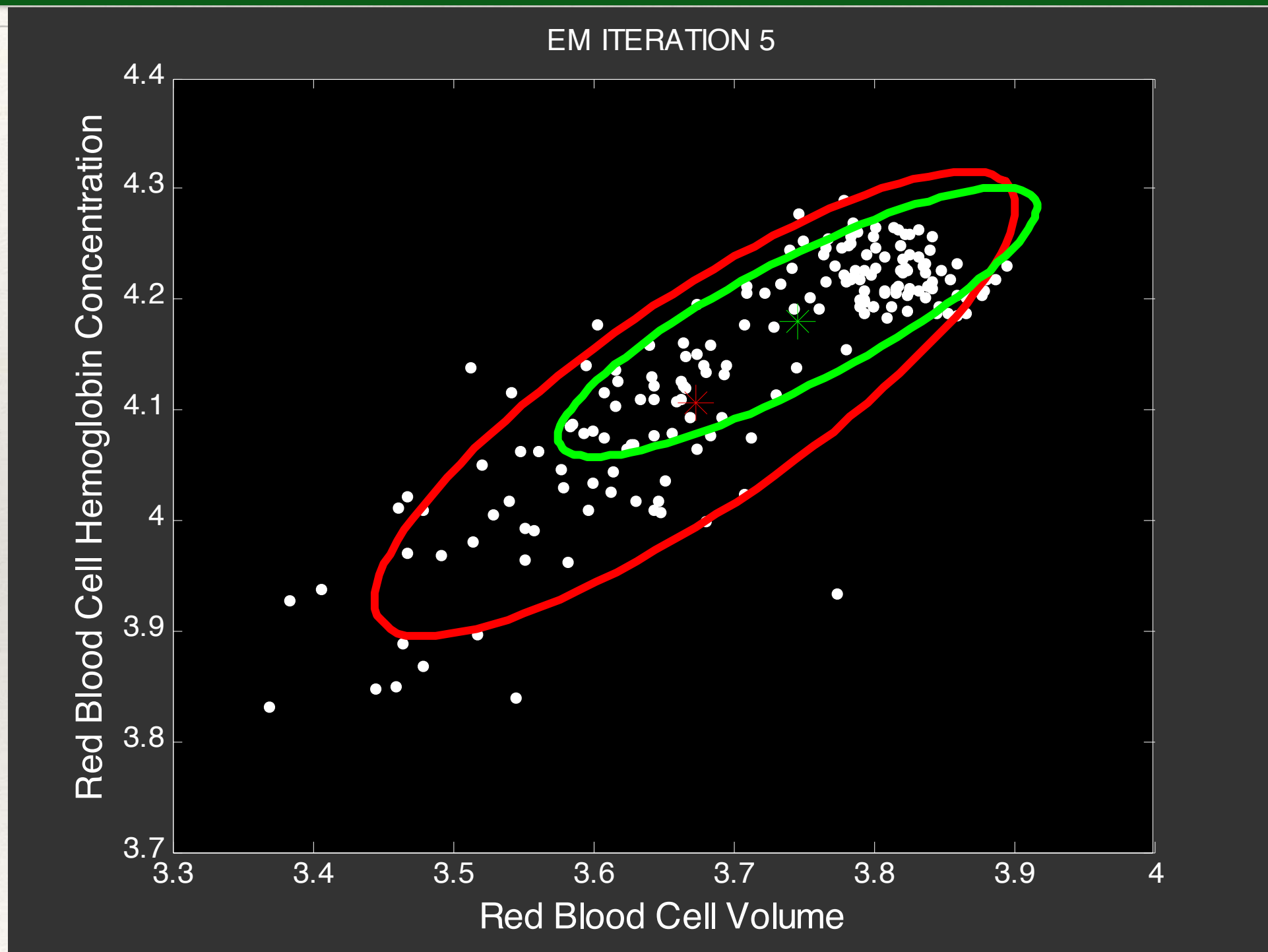
EM Algorithm for GMM



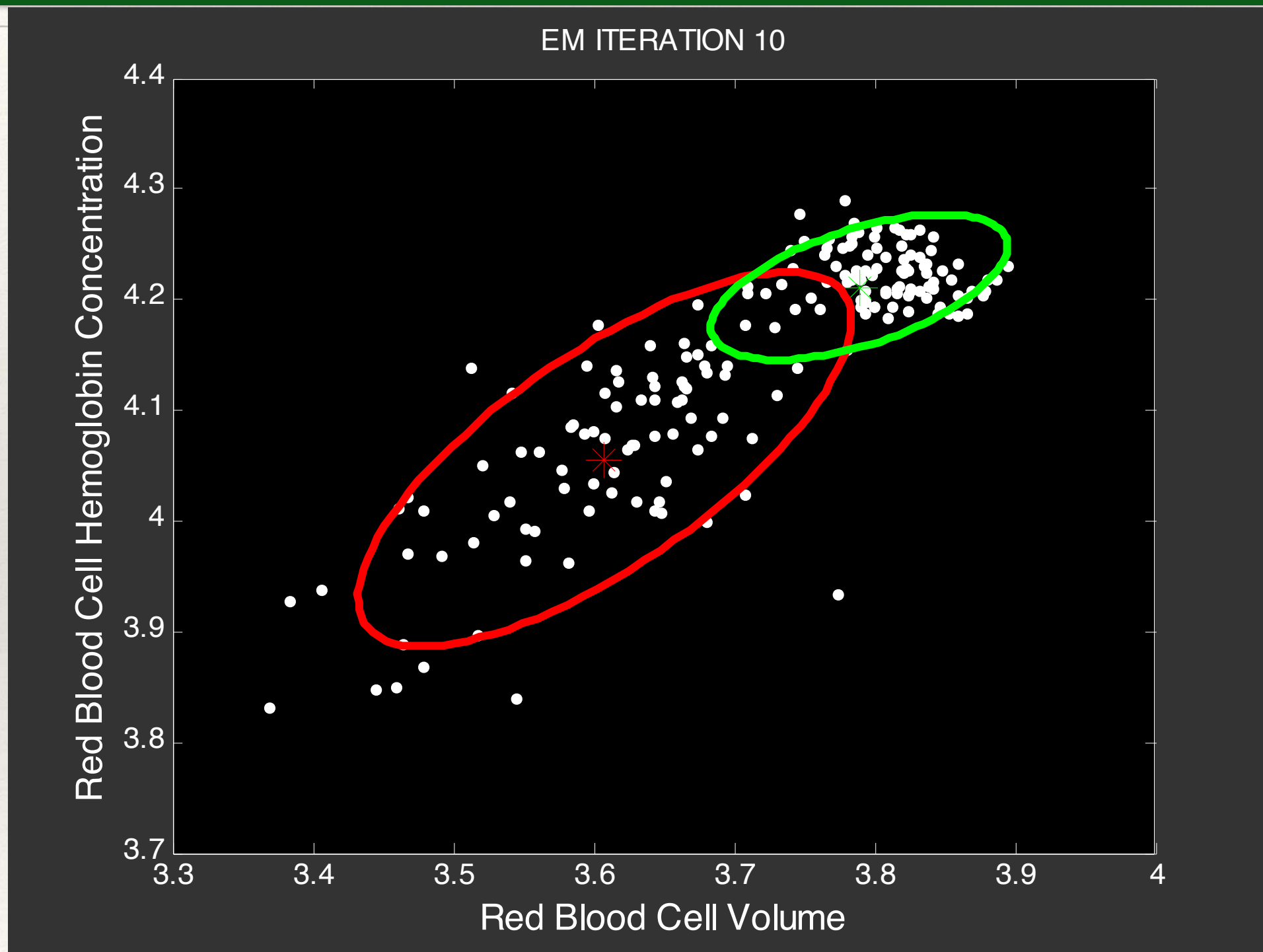
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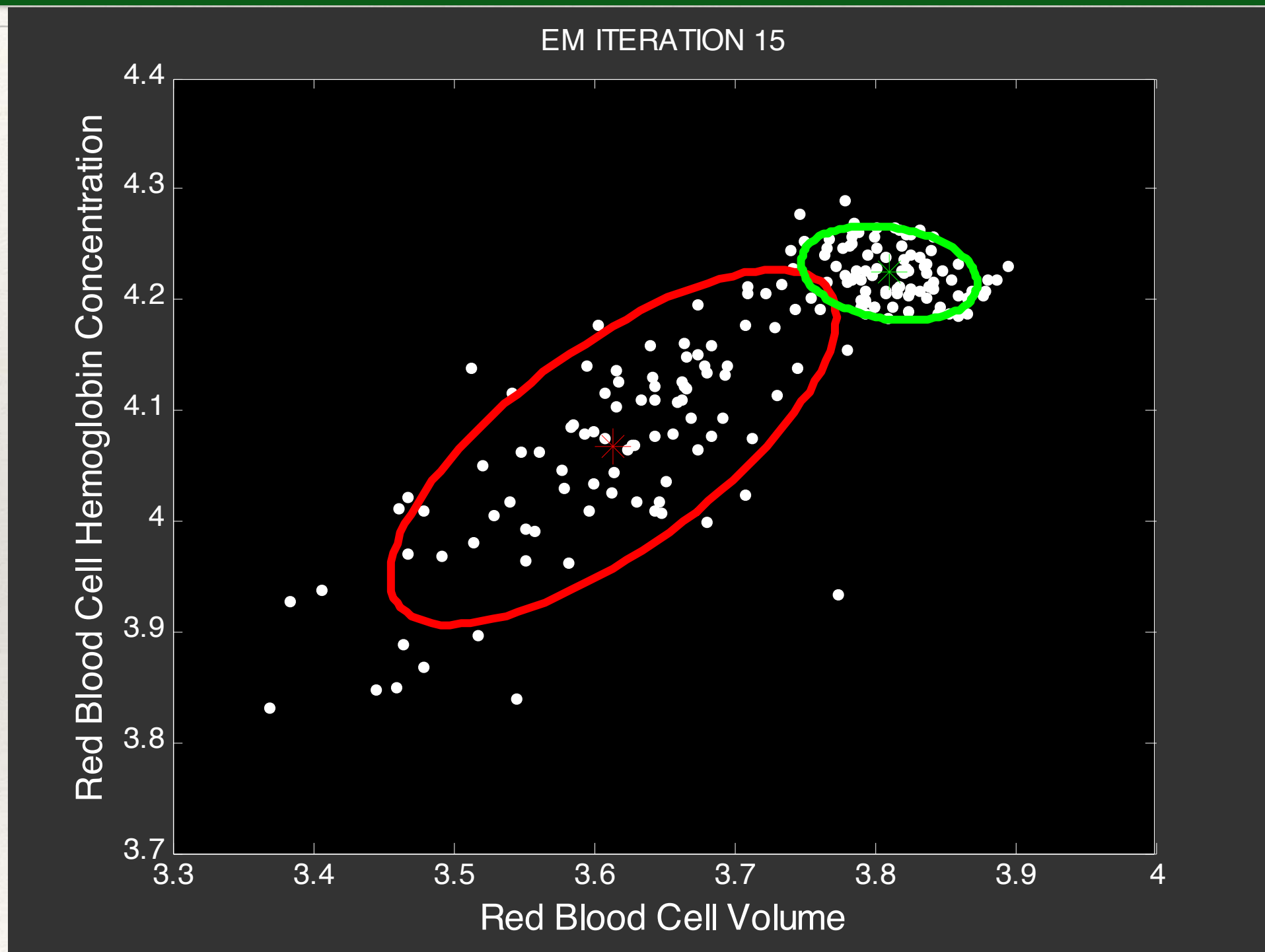
EM Algorithm for GMM



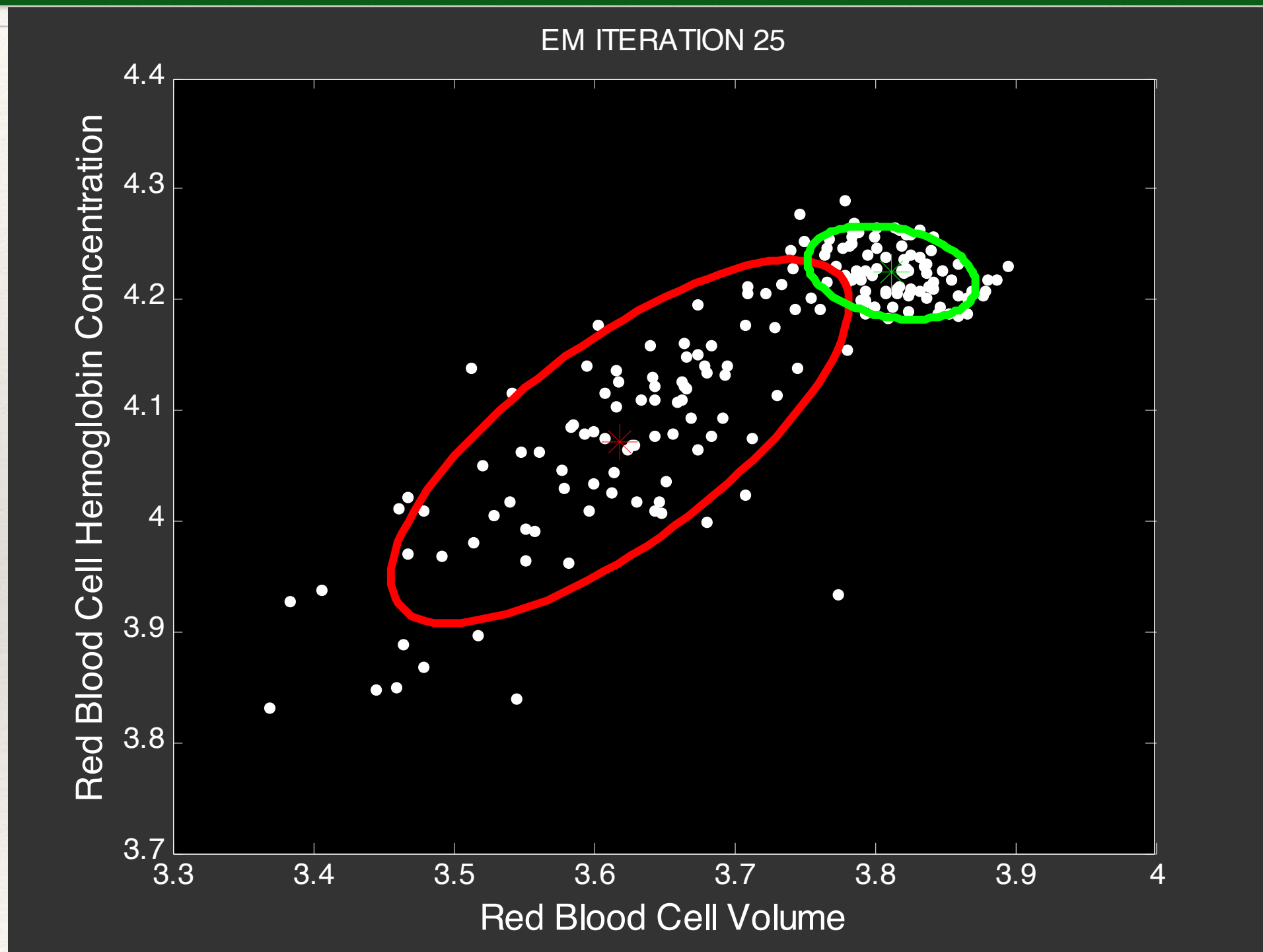
EM Algorithm for GMM



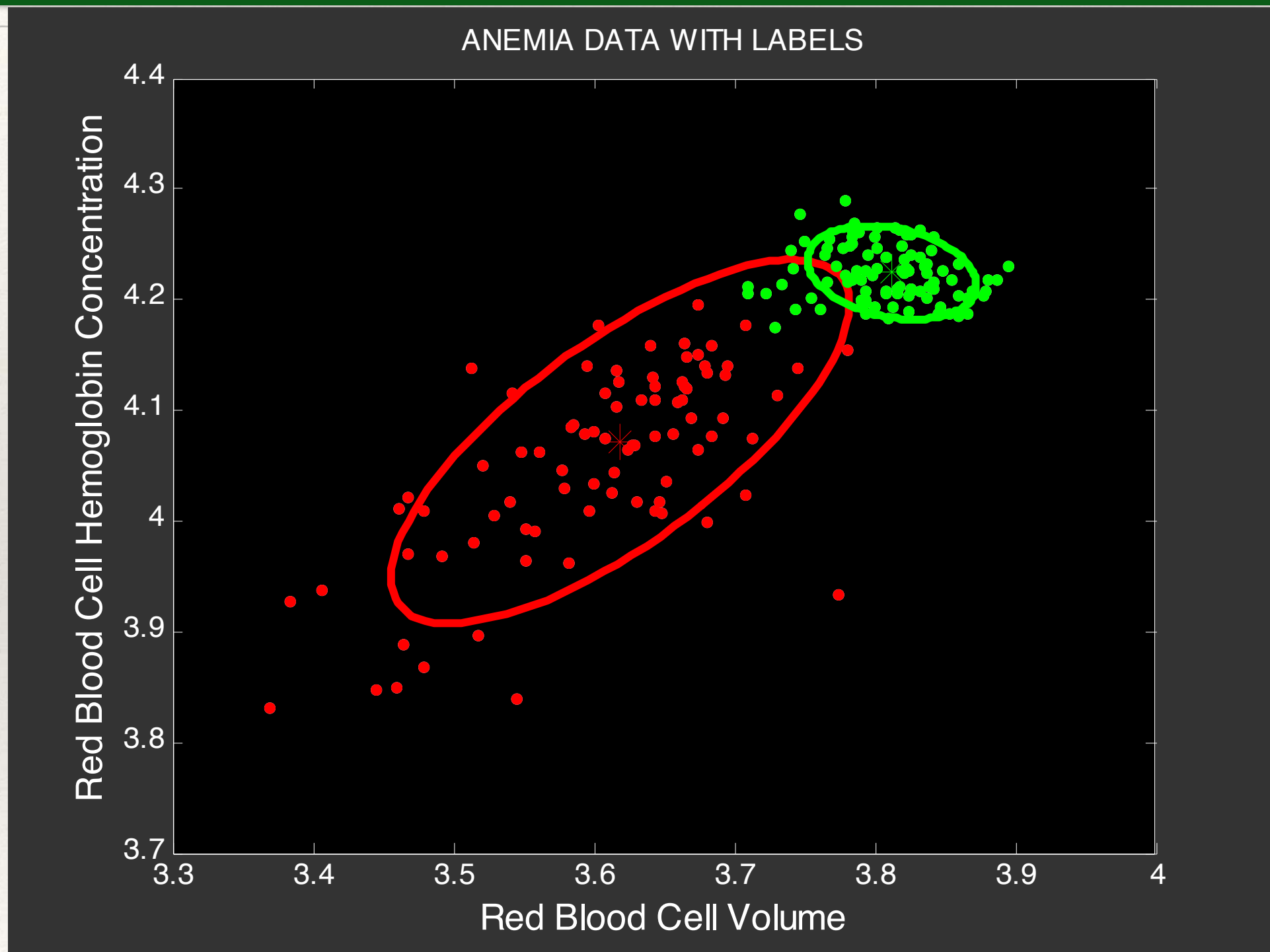
EM Algorithm for GMM



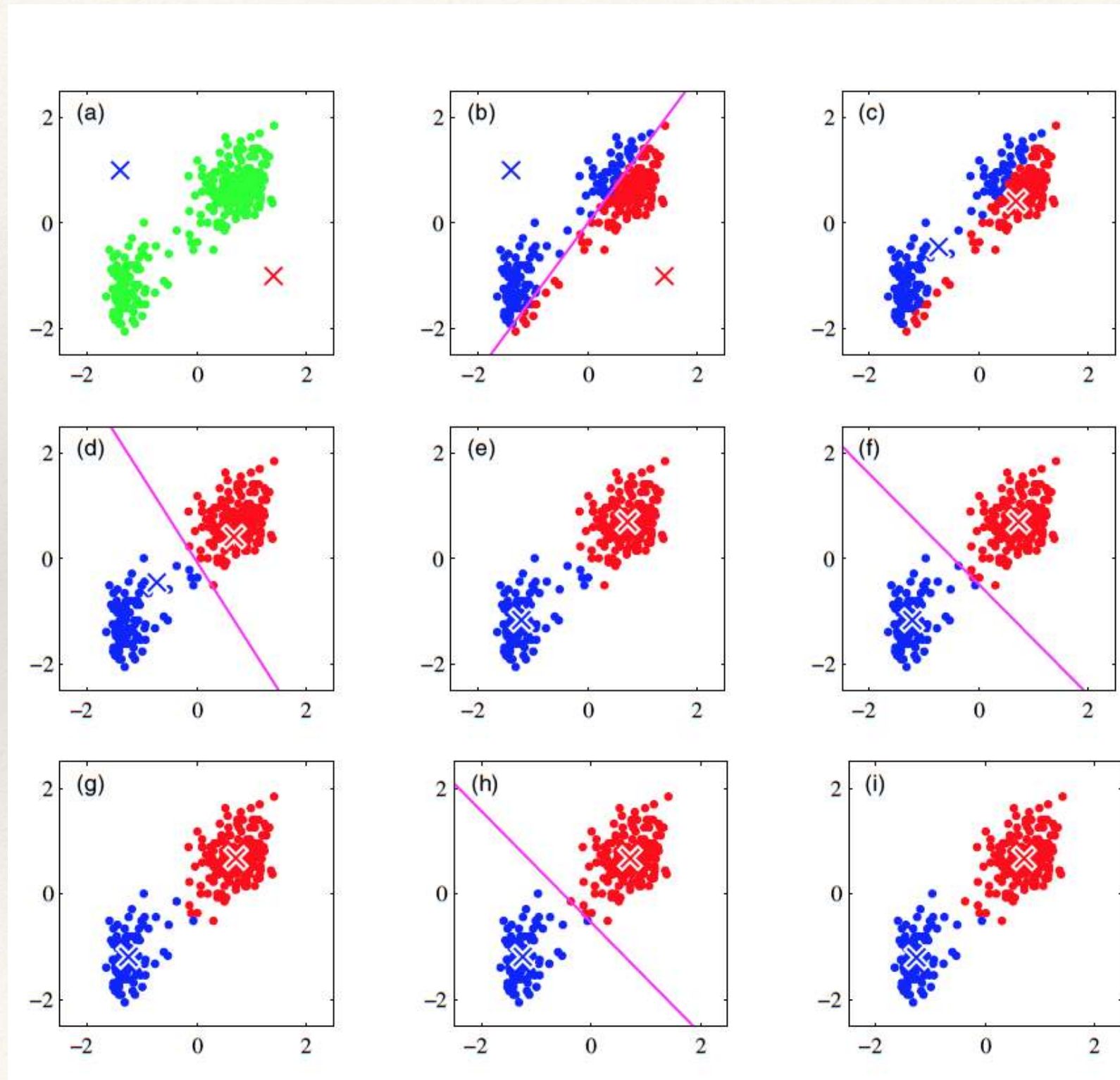
EM Algorithm for GMM



EM Algorithm for GMM

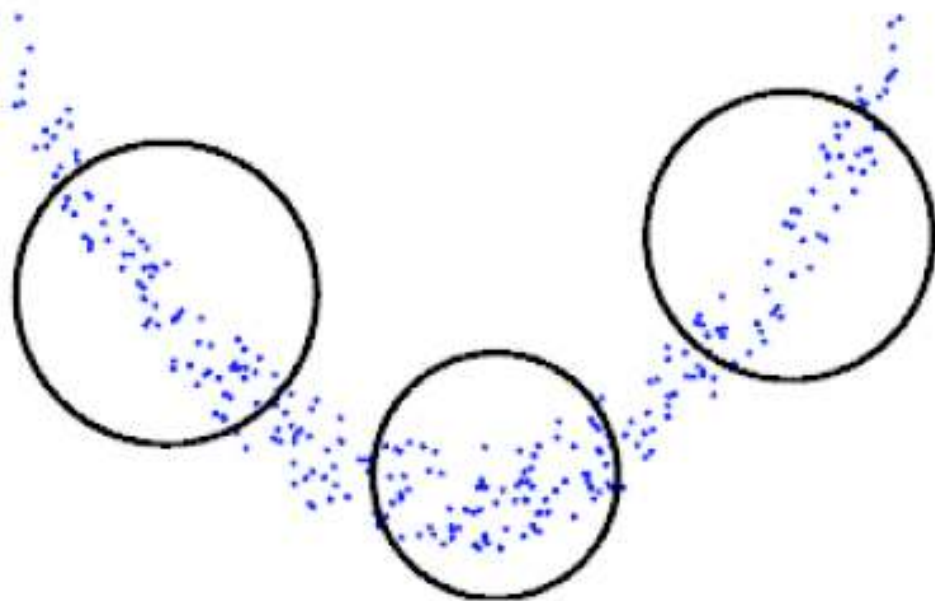


K-means Algorithm for Initialization



Other Considerations

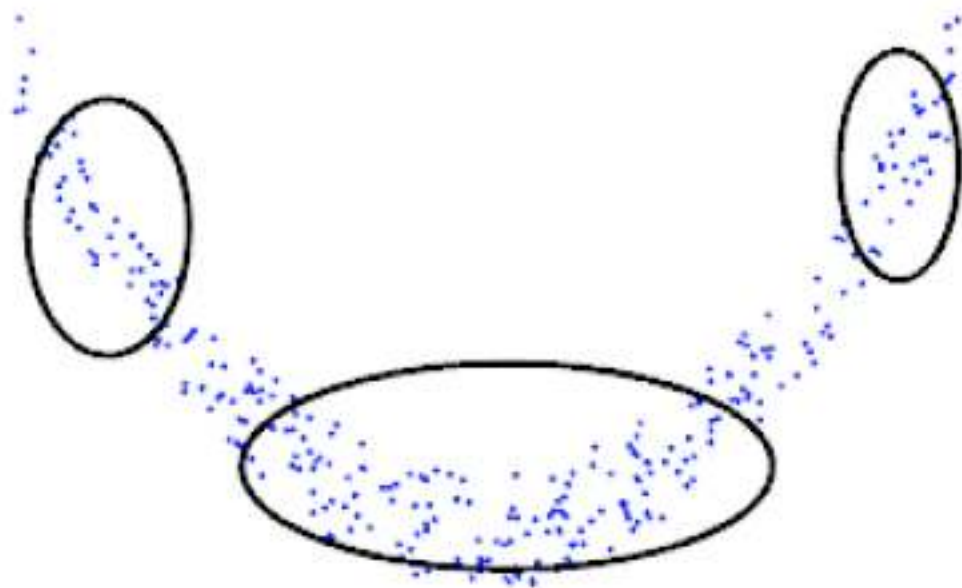
- ❖ Initialization - random or k-means
- ❖ Number of Gaussians
- ❖ Type of Covariance matrix
 - ❖ Spherical covariance



- Less precise.
- Very efficient to compute.

Other Considerations

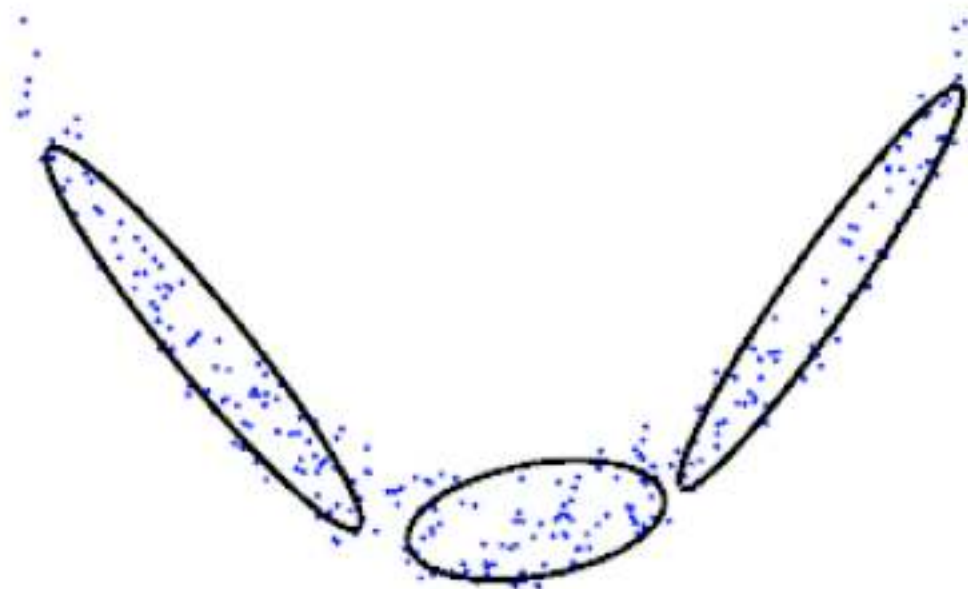
- ❖ Initialization - random or k-means
- ❖ Number of Gaussians
- ❖ Type of Covariance matrix
 - ❖ Diagonal covariance



**-More precise.
-Efficient to compute.**

Other Considerations

- ❖ Initialization - random or k-means
- ❖ Number of Gaussians
- ❖ Type of Covariance matrix
 - ❖ Full covariance



- Very precise.
- Less efficient to compute.